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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

A Project Report on

“AI-Enhanced Dress Sizing and Personalized Size Recommendation System with Privacy Protection”

Done by

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AI-Enhanced Dress Sizing and Personalized Size Recommendation System with Privacy Protection

I. ABSTRACT

This project is about user's recommendation on cloth fittings. The majority of consumers don't receive the precise fit they were hoping for. Thus, this study presents a mechanism that allows the AI to precisely gauge the dimensions of user fittings. The goal of this study is to create a machine learning model that uses a camera-based method to measure dress sizes accurately for both men and women. We all need a tailor in real life to measure our clothes, but we can't do it online, which is why this tool is useful. After measuring, AR comes into play which allows the user to personalize their fittings based on their own choice. User privacy is the main priority, based on their model an avatar is generated once they are measured, allowing them to personalize clothes with their digitally generated image and prevent their faces and bodies from being misused. The project incorporates restrictions against undressing abilities in order to address issues linked to inappropriate AI behaviour. Include a style recommendation system that makes dress recommendations based on the user's personal taste in fashion and current trends. Provide a virtual try-on function so that customers may view several dresses on their avatars before deciding to buy them. Provide a function that assesses the user's body form and gives individualized recommendations for attractive clothing combinations.

II. INTRODUCTION

According to Statista data from 2021, there were over 185 million Indians who shopped online annually. This was a noteworthy rise in comparison to the preceding year. By 2027, 425 million shoppers are projected to be in the country.

Number of annual online shoppers in India from 2019 to 2021, with an estimate for 2027
(in millions)

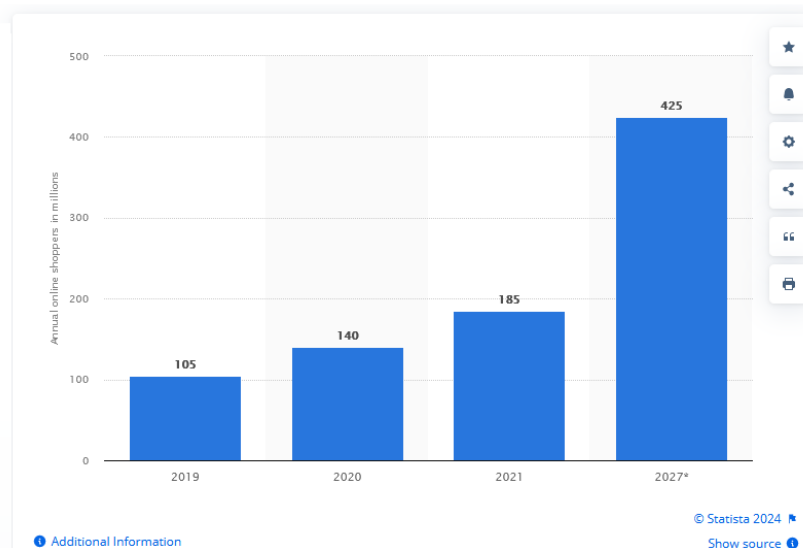


Fig. 2.1: Number of annual online shoppers in India

India is the world's second-largest internet market, with over 800 million users and 125.94 lakh UPI transactions in 2022. In India, almost all pin codes have experienced the advent of e-commerce. In India, tier two cities and smaller towns account for over 60% of transactions and orders.

India is expected to generate US\$104,931.8 million in revenue by 2024, making it the seventh-largest e-commerce market, surpassing Indonesia. Compound annual growth rate (CAGR 2024–2028) of 17.4% is anticipated in revenue, translating into a predicted market volume of US\$199,256.1 million by 2028.

Clothes and shoes are the two most popular online purchase categories among Indian customers. In our representative online poll, 51% and 38% percent of customers, respectively, selected these responses.

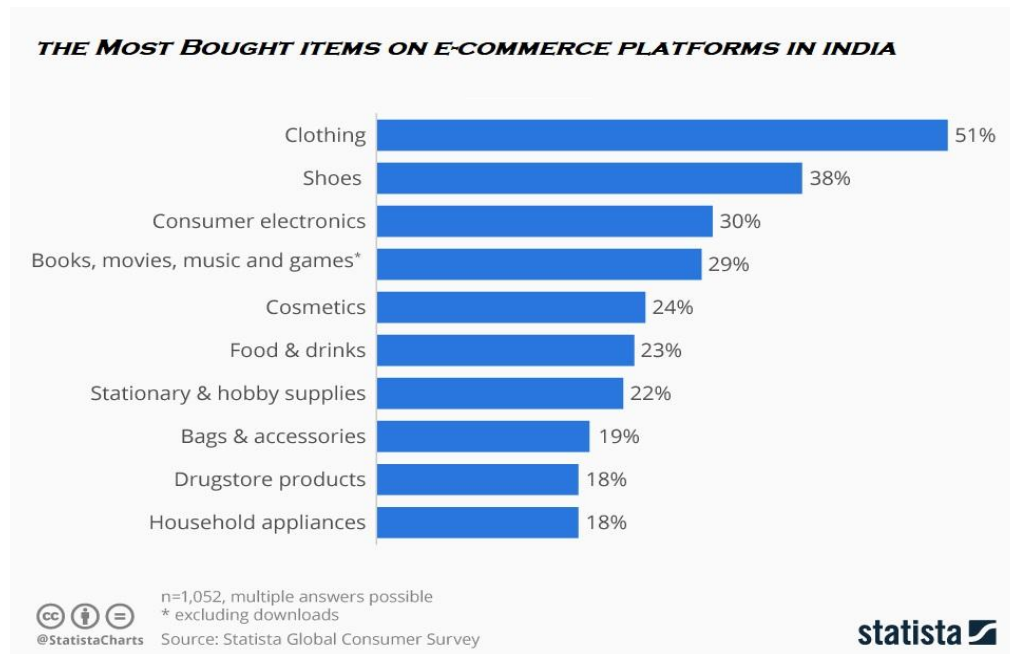


Fig. 2.2: The Most bought items on e-commerce platforms in India

Similarly, when asked which category accounted for the majority of returned online purchases, the majority of Indian respondents stated "clothing." In our online survey from 2023, 47% stated they did so.

Which types of products have you returned in the past?

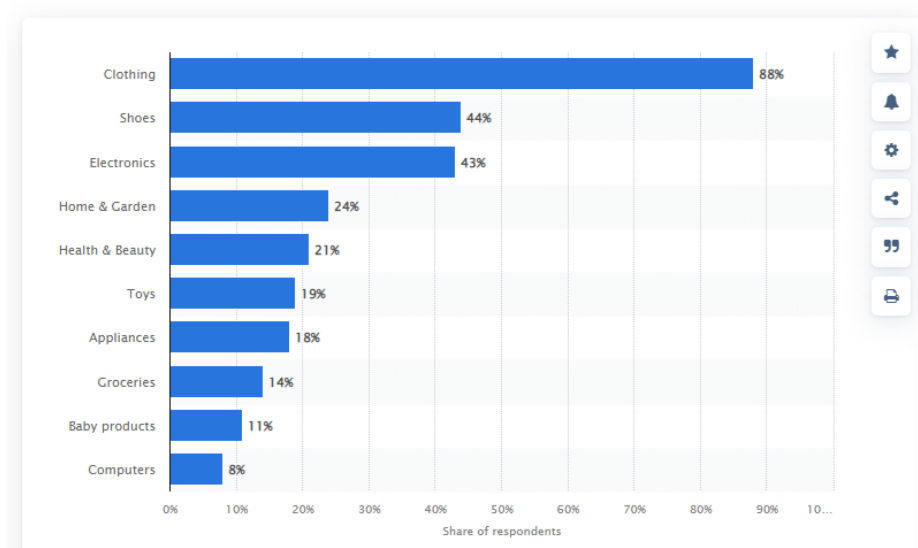


Fig. 2.3: The Most returned items on e-commerce platforms in India

This issue brought us to do this project. In the realm of online shopping, achieving the perfect fit remains a significant challenge for consumers. Addressing this issue, our project focuses on revolutionizing the clothing fitting experience through innovative AI technology. We aim to develop a machine learning model capable of accurately measuring dress sizes using a camera-based method, catering to both men and women. Unlike traditional tailoring, this tool offers online users the convenience of precise fittings from the comfort of their homes. Privacy concerns are paramount; hence, we prioritize safeguarding user data by generating avatars for personalized fittings, preventing misuse of facial and bodily features. Our project integrates AI limitations to prevent inappropriate behaviour and incorporates a style recommendation system, considering individual fashion preferences and current trends. Additionally, a virtual try-on feature empowers customers to visualize various garments on their avatars before making purchase decisions. By assessing body shapes, our system provides tailored clothing combinations, enhancing the overall online shopping experience.



III. OBJECTIVES

The primary objective of this project is to develop an AI-driven mechanism for precisely measuring clothing sizes through a camera-based method, catering to both men and women. The aim is to address the prevalent issue of ill-fitting garments in online shopping by providing users with accurate measurements and personalized fittings. Additionally, the project seeks to prioritize user privacy by generating avatars based on measurements, ensuring protection against misuse of personal data. Furthermore, the project aims to incorporate AI limitations to prevent inappropriate behaviour and integrate a style recommendation system that considers individual fashion preferences and current trends. Finally, the project aims to enhance the online shopping experience by providing a virtual try-on feature and offering personalized clothing combinations based on the user's body shape.

IV. SYSTEM REQUIREMENTS

4.1: HARDWARE REQUIREMENTS:

1. Basic Camera
2. Laptop
3. Storage Device
4. Graphics Processing Unit (GPU)

4.2: SOFTWARE REQUIREMENTS:

1. Windows OS.
2. Visual Studio Code Software.
3. Python 3.7.0.
4. MySQL

5. GitHub
6. Image Processing Libraries (OpenCV, Media pipe)
7. Pose Estimating Libraries

V. METHODOLOGY

The prototype is developed in three modules as follows:

- **Module I:** User Side Interface
- **Module II:** Measuring Unit
- **Module III:** Virtual Try-on Unit
- **Module IV:** Avatar Unit

5.1: Module I: User Side Interface

The user interface (UI) for this project should be intuitive, user-friendly and accessible across various devices, including smartphones, tablets and computers. User side unit is where user can choose their cloth to be bought.

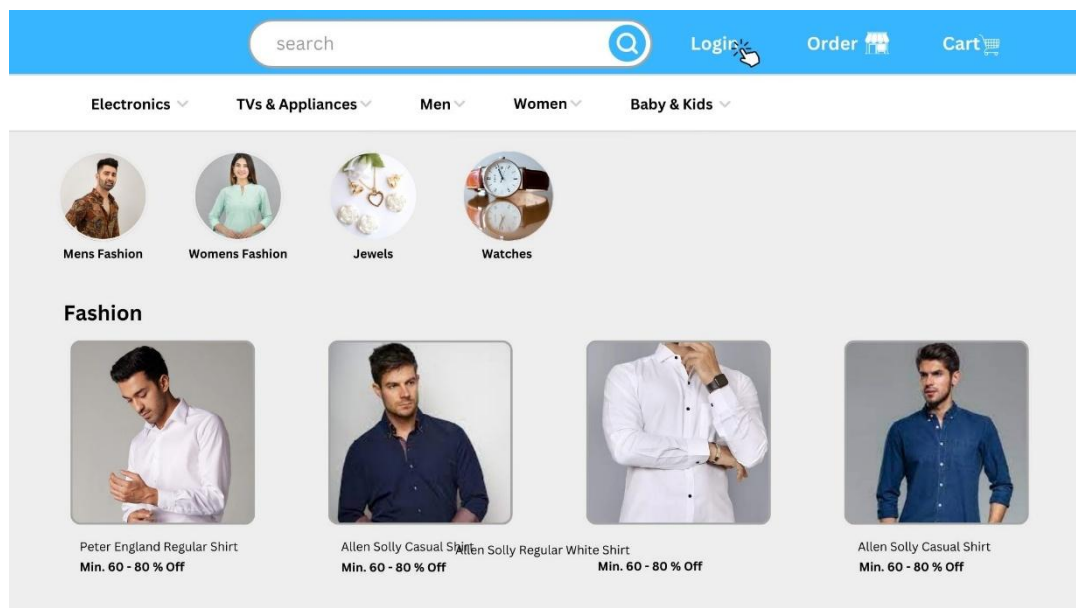


Fig. 5.1.1. User Interface

Users should be able to create an account or log in securely to access personalized features such as measurement history, saved preferences and recommendations.

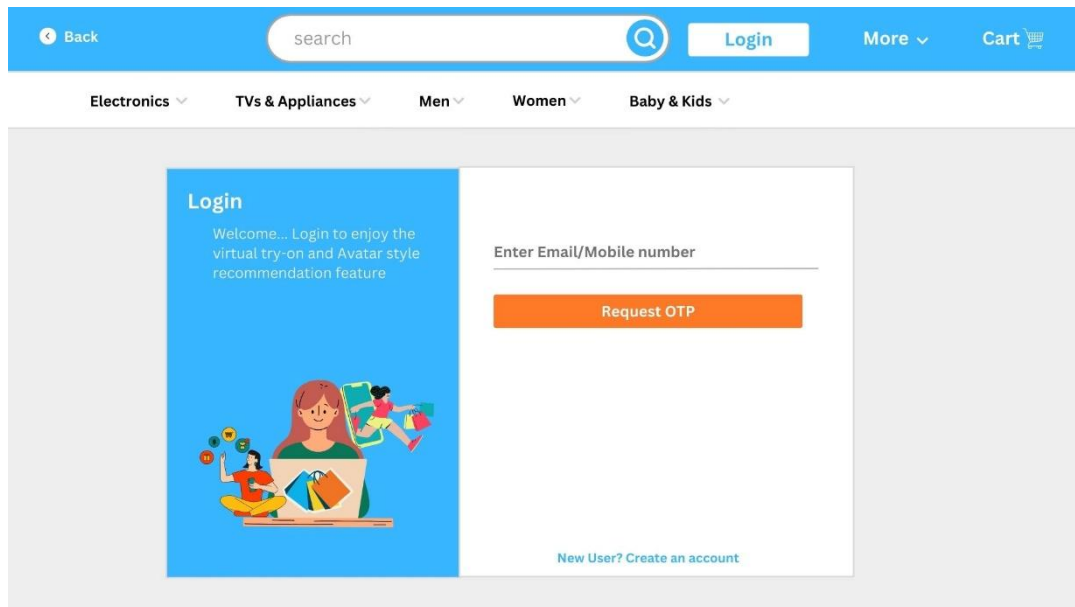


Fig. 5.1.2. User Login in with Mobile Number and Email

Users will receive an OTP after logging in with their email address or mobile number as shown in Fig.5.1.2. After the user enters the OTP, an account has been created for them, and they are successfully signed in as shown in Fig.5.1.3.

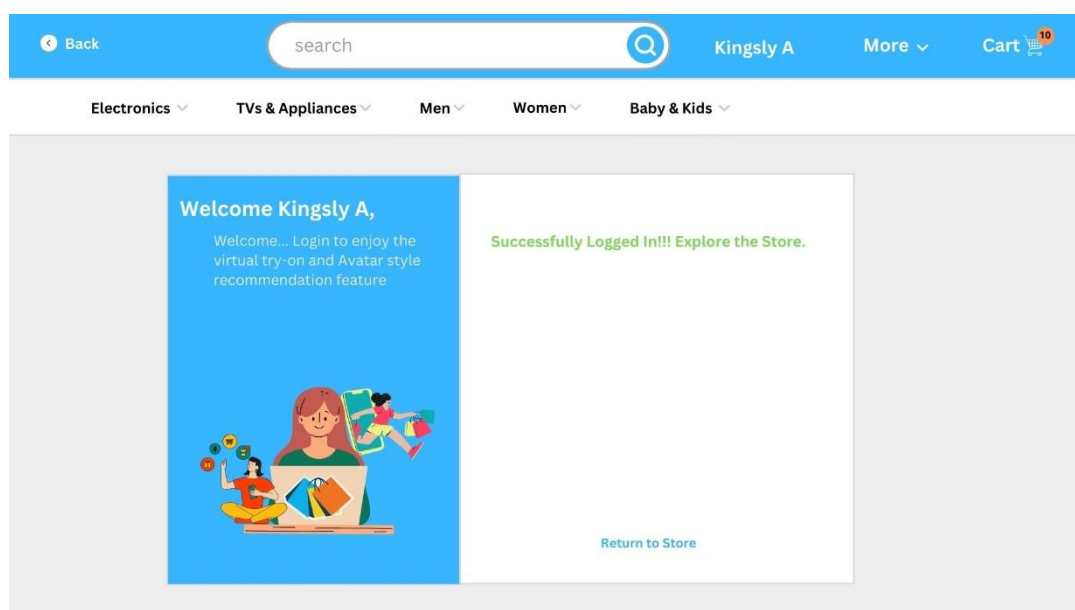


Fig. 5.1.3. User Successfully signed in.

5.2: Module II: Measuring Unit

The user can select which cloth to purchase after logging in as shown in Fig.5.2.1.

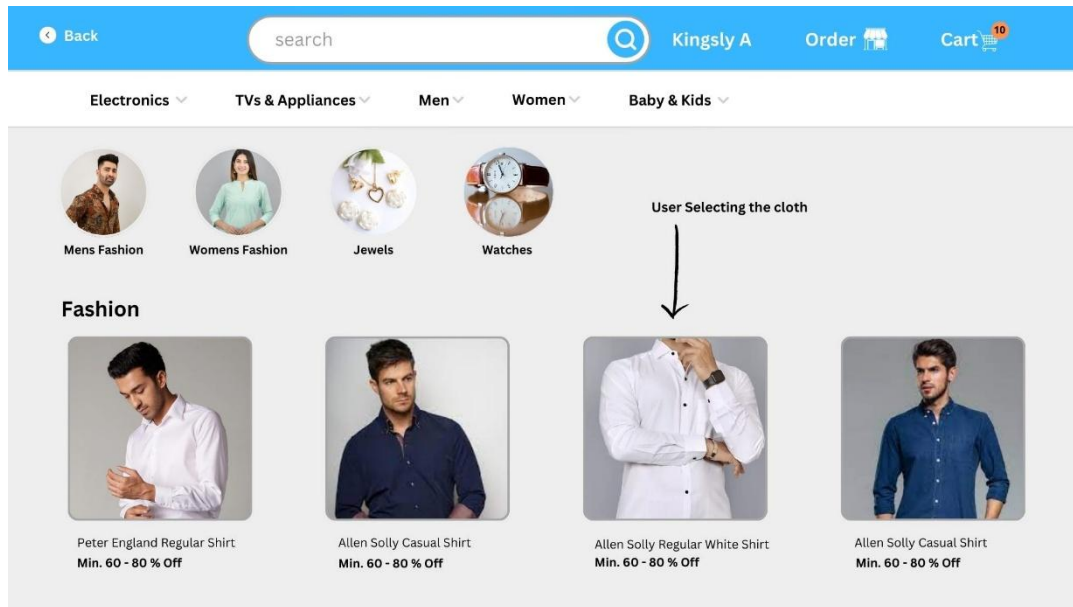


Fig. 5.2.1. User Selecting the cloth.

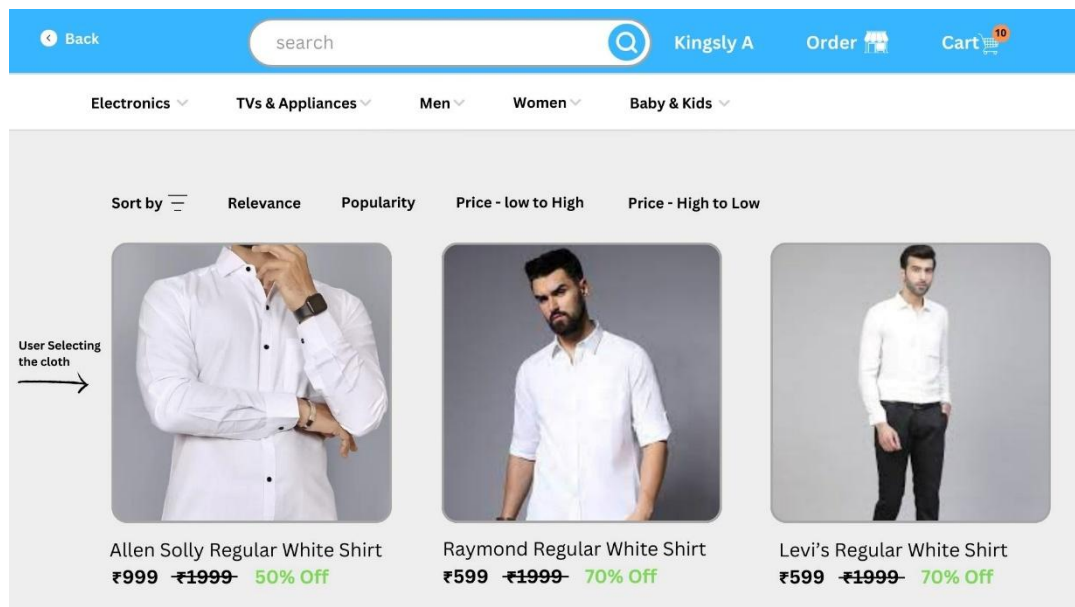


Fig. 5.2.2. User Selecting the cloth which they want to buy.

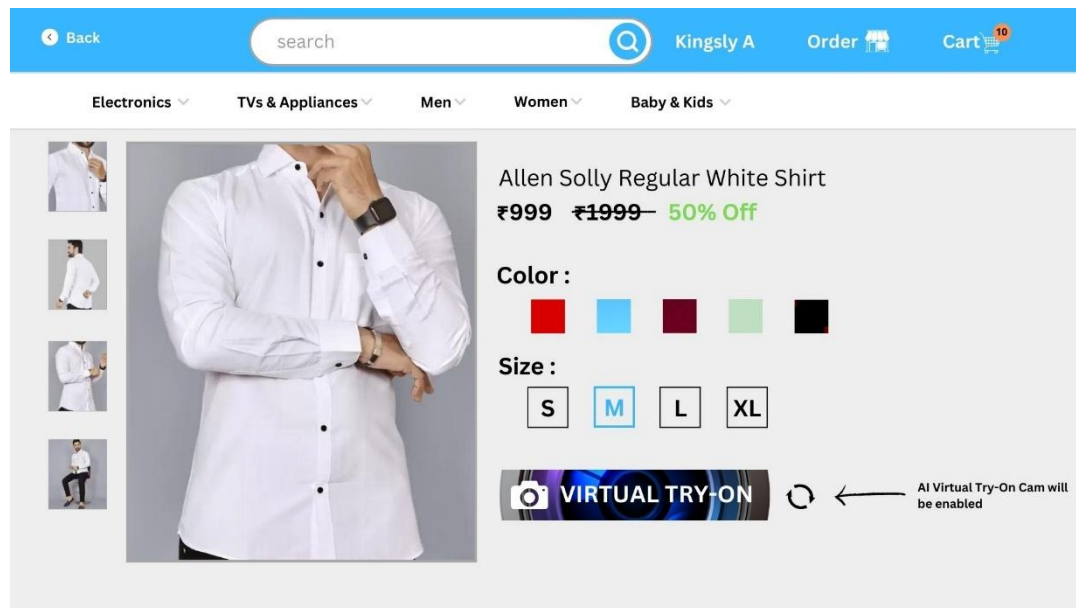


Fig. 5.2.3. AI Virtual Try-On Camera will be enabled to user.

Options like as color, size, and an AI virtual try-on camera will appear once the consumer has chosen the cloth to purchase as shown in Fig.5.2.3.

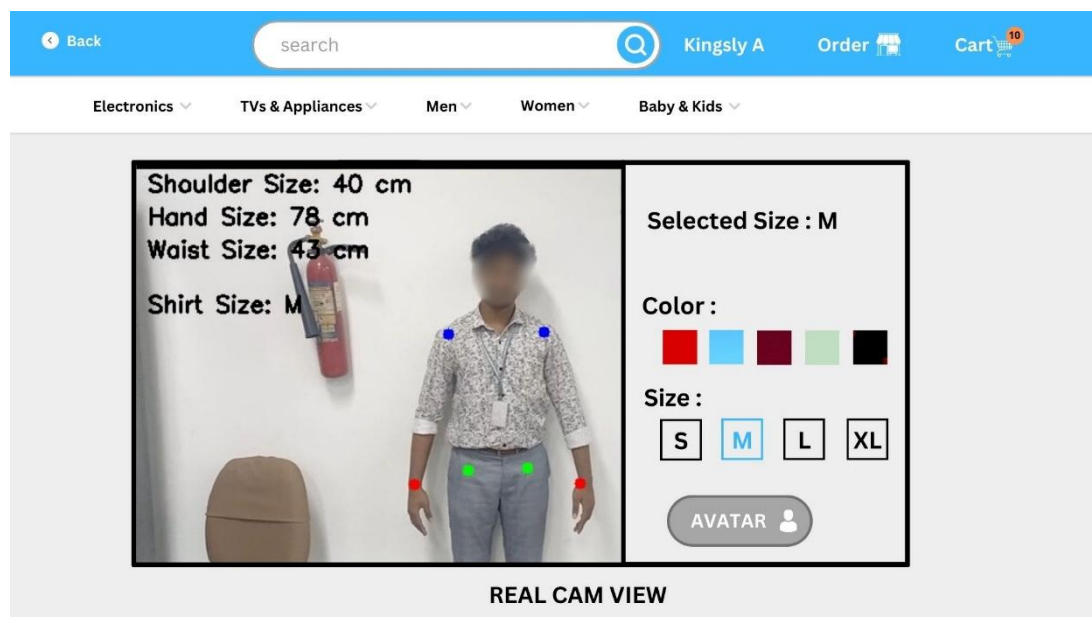


Fig. 5.2.4. AI Virtual Try-On Camera will be enabled to user.

When the user enables the AI virtual try-on camera, it measures the customer's waist, hands, and shoulder. Based on those measurements, it predicts and suggests the appropriate clothing size, such as S, M, L, XL, or XXL as shown in Fig.5.2.4.

5.3: Module III: Virtual Try-on Unit

Once a cloth size is predicted, the Virtual Try-On Unit helps the user to virtually put the clothing on and determines whether it will fit for their skin tone, hair type, and body type.

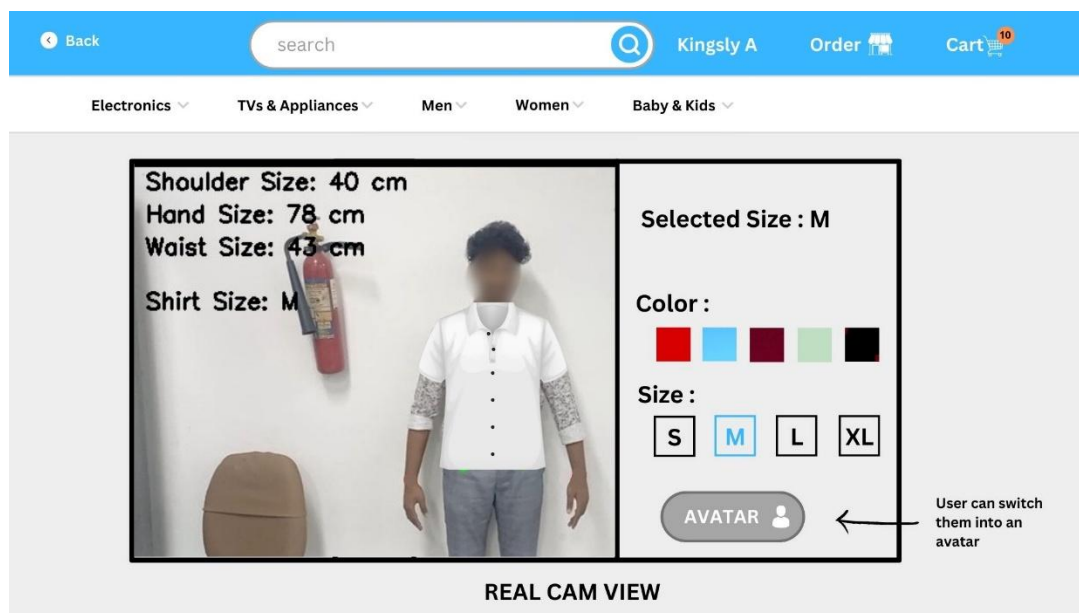


Fig. 5.3.1. Virtual Try-On Unit overlays the selected cloth.

Computer vision algorithms examine the user's body shape and proportions to assess how the clothing fits. This includes checking for tightness, looseness, length, and overall suitability as shown in Fig.5.3.1.

5.4: Module IV: Avatar Unit

Using AR technology, the Avatar Unit overlays the selected clothing onto a 3D avatar model of the user. This allows the user to see how the clothing looks on them in real-time. It is also viewed in the 360 degree view.

With the help of Avatar Unit's options for customization, users can instantly change the fit of the clothes to better fit their tastes. They can also add accessories and footwear, such as a watch, chain, cap, earrings, shoes, slippers, or none at all.

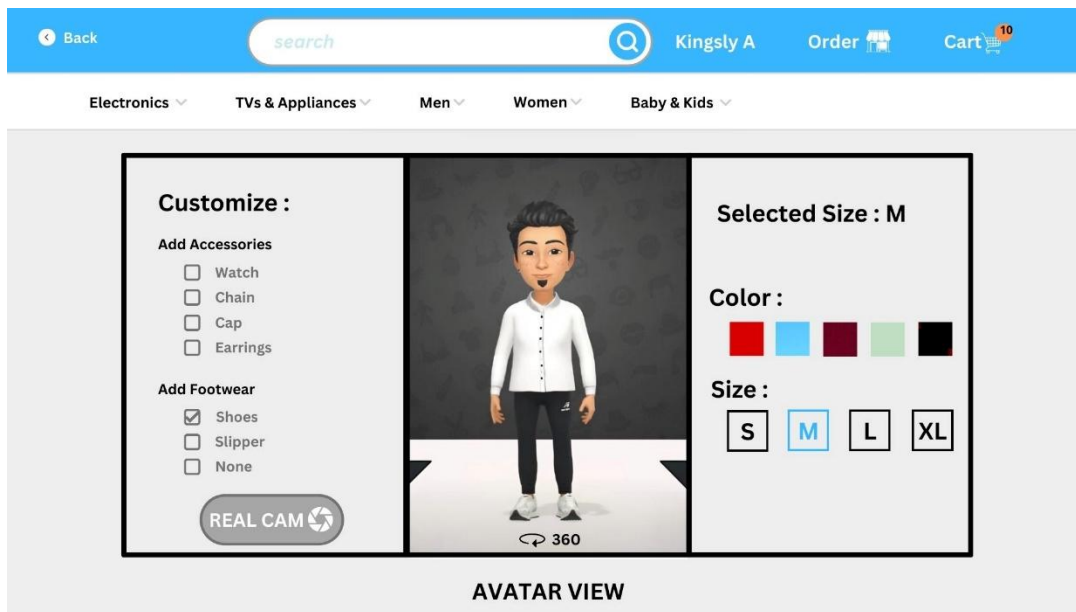


Fig. 5.4.1. Avatar Unit overlays the selected cloth.

VI. CONCLUSION

This project finds accurate dress sizing prediction and customized size suggestions, customers will be better equipped for making purchases, which will decrease returns and boost customer happiness. Through analysis of multiple data points, including body dimensions, fit preferences, and style preferences, the technology offers a customized purchasing experience for every user.